# Ridge Regression, Gradient Descent, And Perceptron

## CSAI 253 - Spring 2024

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# Revision of Standard Linear Regression Model

3

*pending recreation*

4

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# Problems of Linear Regression

*(With Linearly Separable Data)*

# Problems of Linear Regression

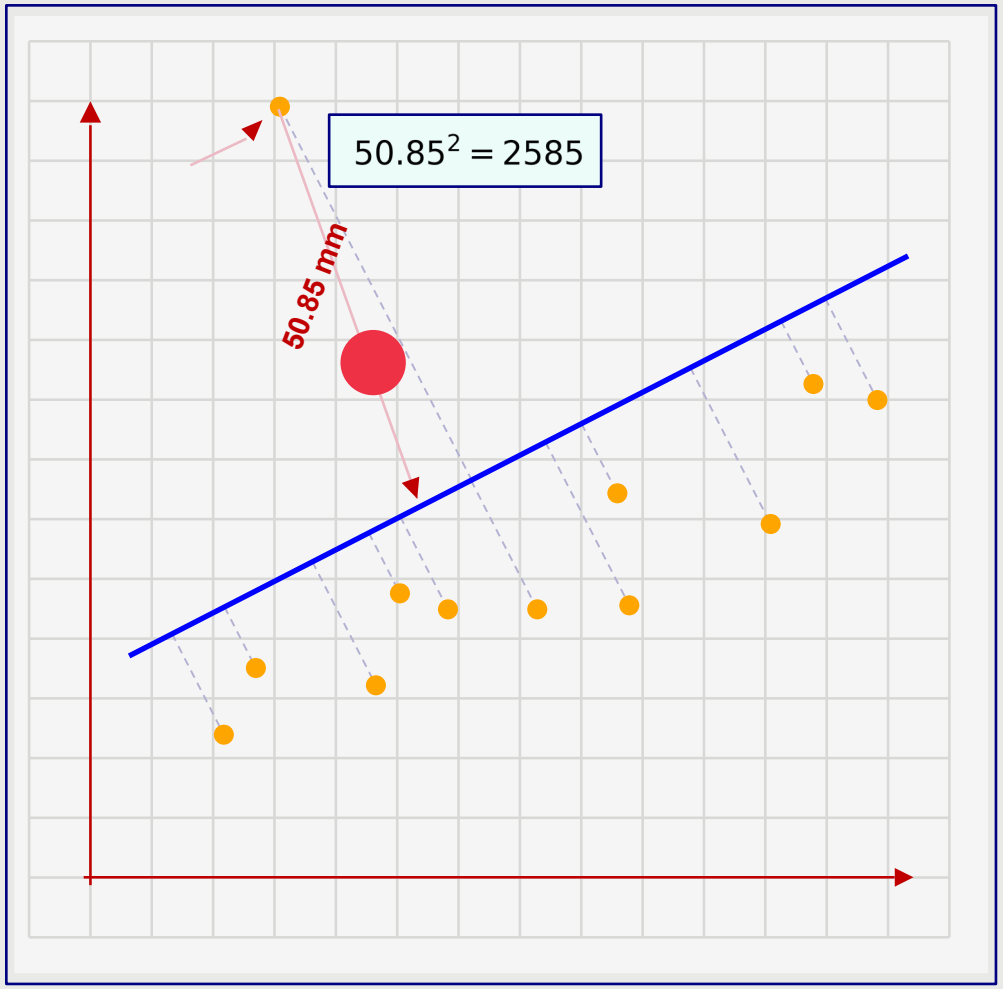
Assume the Dataset is linearly separable (suitable to apply Linear regression model)

**Linear regression model has FOUR problems:**

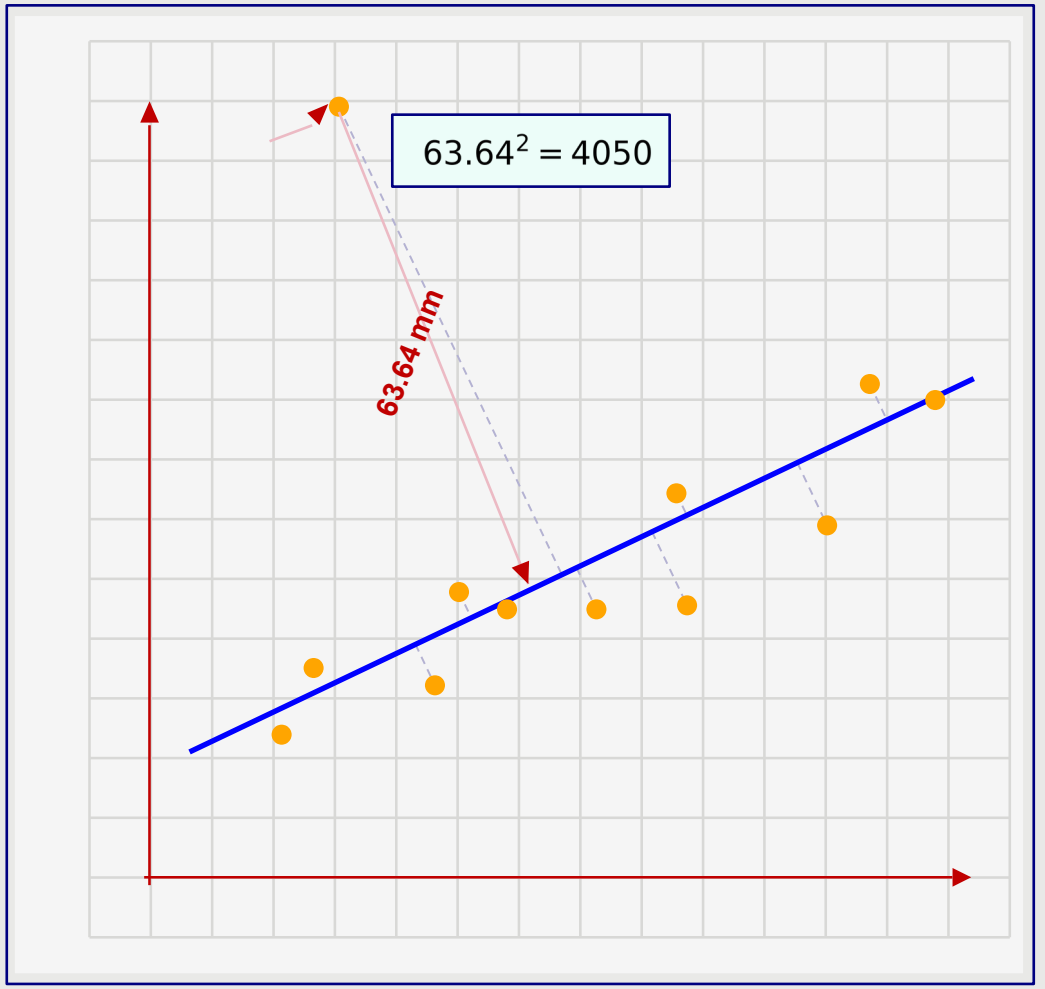
- **Overfitting Problem**
- **Matrix Inversion Problem [1]** ***(Singular Matrix)***
- **Matrix Inversion Problem [2]** ***(Large Matrix dimension)***
- **Different Hyperplanes can solve the problems** ***(Selection of Best Hyperplane)***

# [1] Ridge Regularizer to Minimize Overfitting

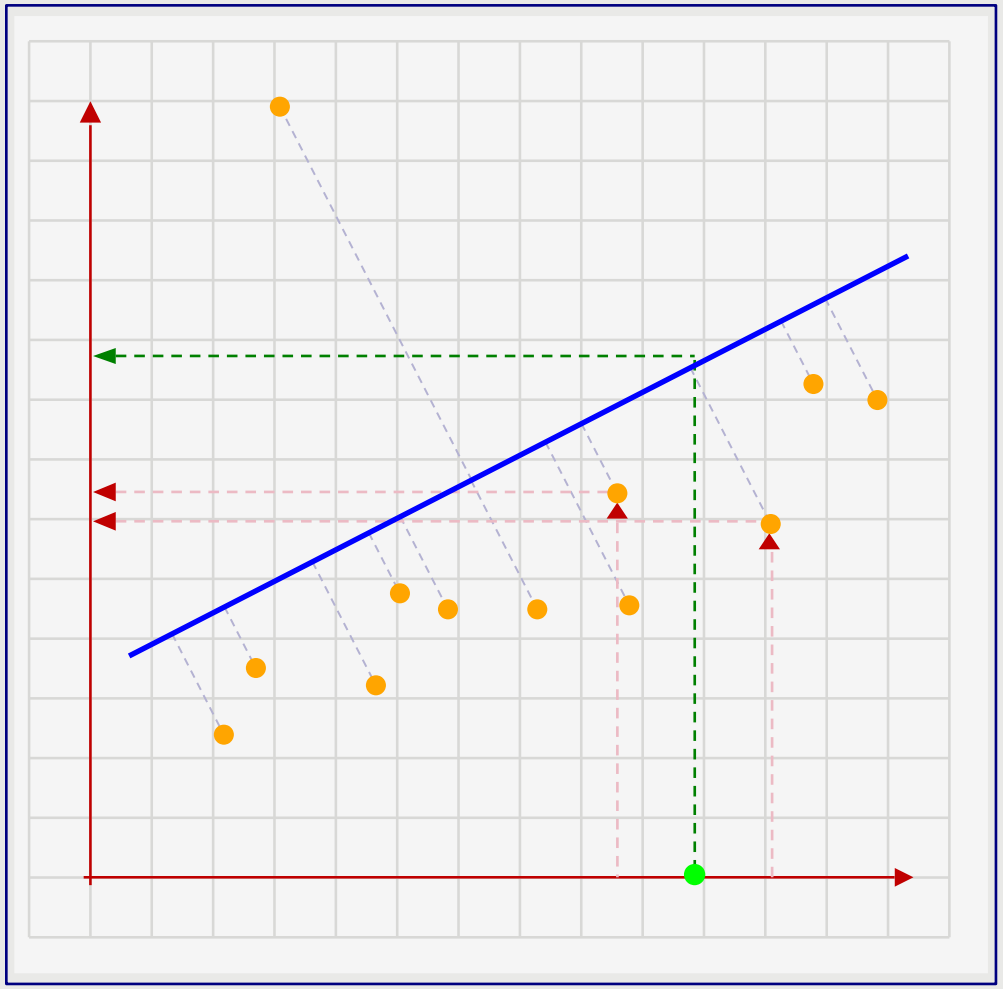
*(Also Known as L2 regularizer)*



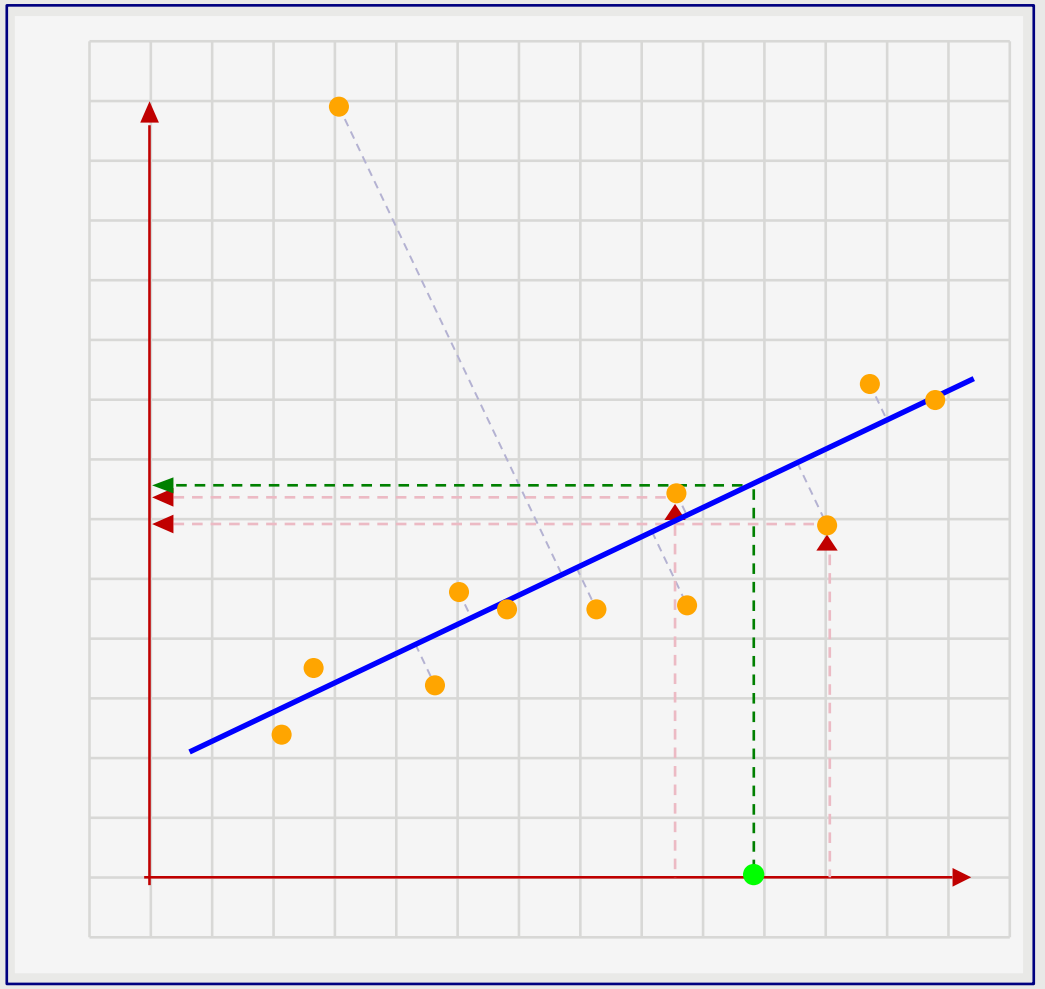

**Overfit** (try to minimize the residuals ONLY)
Fails to detect the Pattern




**Generalized** (Residuals are not minimized)
Successes to detect the Pattern



**Overfit** (try to minimize the residuals ONLY)
Fails to detect the Pattern



**Generalized** (Residuals are not minimized)
Successes to detect the Pattern

# Example of Overfitting due to outlier in data

*(and how to minimize overfitting )*

# Effect of Overfitting on model parameter "W"

**Training Data**

**Perfect Data**
**Y follows x1**

| | X1 | X2 | X3 | X4 | X5 | X6 | X7 | X8 |
|---|---|---|---|---|---|---|---|---|
| X1 | 1 | 2 | 3 | 4 | 8 | 9 | 10 | 11 |
| X2 | 1 | 3 | 5 | 6 | 16 | 7 | 5 | 3 |
| y | 2 | 4 | 6 | 8 | 16 | 18 | 20 | 22 |

**X5 is Outlier**

| | X1 | X2 | X3 | X4 | X5 | X6 | X7 | X8 |
|---|---|---|---|---|---|---|---|---|
| X1 | 1 | 2 | 3 | 4 | 8 | 9 | 10 | 30 |
| X2 | 1 | 3 | 5 | 6 | 16 | 7 | 5 | 3 |
| y | 2 | 4 | 6 | 8 | 16 | 18 | 20 | 22 |

**X5 is Outlier**

| | X1 | X2 | X3 | X4 | X5 | X6 | X7 | X8 |
|---|---|---|---|---|---|---|---|---|
| X1 | 1 | 2 | 3 | 4 | 8 | 9 | 10 | 60 |
| X2 | 1 | 3 | 5 | 6 | 16 | 7 | 5 | 3 |
| y | 2 | 4 | 6 | 8 | 16 | 18 | 20 | 22 |

**Test Data**

| | Test1 | Test2 | Test3 |
|---|---|---|---|
| X1 | 30 | 40 | 50 |
| X2 | 15 | 20 | 25 |
| y | 60 | 80 | 100 |

**Again**
**Y follows x1**

What will be the effect of X8 Outlier values on the trained model ???

# Effect of Overfitting on model parameter "W"



**Train**

| | $X_1$ | $X_2$ | $X_3$ | $X_4$ | $X_5$ | $X_6$ | $X_7$ | $X_8$ |
|---|---|---|---|---|---|---|---|---|
| $X_1$ | 1 | 2 | 3 | 4 | 8 | 9 | 10 | 11 |
| $X_2$ | 1 | 3 | 5 | 6 | 16 | 7 | 5 | 3 |
| y | 2 | 4 | 6 | 8 | 16 | 18 | 20 | 22 |

$$W = \begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix}$$

$$\|W\|^2 = 4$$

**Test**

| $Test_1$ | $Test_2$ | $Test_3$ |
|---|---|---|
| 30 | 40 | 50 |
| 15 | 20 | 25 |
| 60 | 80 | 100 |

| | $X_1$ | $X_2$ | $X_3$ | $X_4$ | $X_5$ | $X_6$ | $X_7$ | $X_8$ |
|---|---|---|---|---|---|---|---|---|
| $X_1$ | 1 | 2 | 3 | 4 | 8 | 9 | 10 | 30 |
| $X_2$ | 1 | 3 | 5 | 6 | 16 | 7 | 5 | 3 |
| y | 2 | 4 | 6 | 8 | 16 | 18 | 20 | 22 |

$$W = \begin{pmatrix} 2.7 \\ 0.67 \\ 0.64 \end{pmatrix}$$

$$\|W\|^2 = 8.11$$

| $Test_1$ | $Test_2$ | $Test_3$ |
|---|---|---|
| 30 | 40 | 50 |
| 15 | 20 | 25 |
| 32.4 | 42.3 | 52.3 |

| | $X_1$ | $X_2$ | $X_3$ | $X_4$ | $X_5$ | $X_6$ | $X_7$ | $X_8$ |
|---|---|---|---|---|---|---|---|---|
| $X_1$ | 1 | 2 | 3 | 4 | 8 | 9 | 10 | 60 |
| $X_2$ | 1 | 3 | 5 | 6 | 16 | 7 | 5 | 3 |
| y | 2 | 4 | 6 | 8 | 16 | 18 | 20 | 22 |

$$W = \begin{pmatrix} 4 \\ 0.29 \\ 0.78 \end{pmatrix}$$

$$\|W\|^2 = 16.7$$

| $Test_1$ | $Test_2$ | $Test_3$ |
|---|---|---|
| 30 | 40 | 50 |
| 15 | 20 | 25 |
| 24.4 | 31.2 | 38 |

*pending recreation*

*pending recreation*

# $L_1$ and $L_2$ Regularization



**Regularizers are based on:**

**Penalize large values of W (model parameters)**

- This have the effect of better generalization of the model
- Since there is a contradiction between model training (using training dataset) and model generalization, the generalization term is added to the objective loss function with a customized weigh factor (do regularization to what extent)
- Penalization may be applied on the Squared Norm of the Weight Vector
  This is called $L_2$ regularization (also know as Ridge Regularization )
- Penalization may also be applied on the sum of Absolute values of the Weight Vector
  This is called $L_1$ regularization
- Regularization is particularly important for Small number of samples (N) and Large vector Length (d)

# Adding Ridge Regularizer to the model λ=3



**Train**

| | $X_1$ | $X_2$ | $X_3$ | $X_4$ | $X_5$ | $X_6$ | $X_7$ | $X_8$ |
|---|---|---|---|---|---|---|---|---|
| $X_1$ | 1 | 2 | 3 | 4 | 8 | 9 | 10 | 11 |
| $X_2$ | 1 | 3 | 5 | 6 | 16 | 7 | 5 | 3 |
| y | 2 | 4 | 6 | 8 | 16 | 18 | 20 | 22 |

$$W = \begin{pmatrix} 0.1 \\ 1.95 \\ 0.03 \end{pmatrix}$$

$$\|W\|^2 = 3.8$$

**Test**

| | $Test_1$ | $Test_2$ | $Test_3$ |
|---|---|---|---|
| $X_1$ | 30 | 40 | 50 |
| $X_2$ | 15 | 20 | 25 |
| y | 59 | 78.7 | 98.4 |

| | $X_1$ | $X_2$ | $X_3$ | $X_4$ | $X_5$ | $X_6$ | $X_7$ | $X_8$ |
|---|---|---|---|---|---|---|---|---|
| $X_1$ | 1 | 2 | 3 | 4 | 8 | 9 | 10 | 30 |
| $X_2$ | 1 | 3 | 5 | 6 | 16 | 7 | 5 | 3 |
| y | 2 | 4 | 6 | 8 | 16 | 18 | 20 | 22 |

$$W = \begin{pmatrix} 1.2 \\ 0.72 \\ 0.77 \end{pmatrix}$$

$$\|W\|^2 = 2.45$$

| | $Test_1$ | $Test_2$ | $Test_3$ |
|---|---|---|---|
| $X_1$ | 30 | 40 | 50 |
| $X_2$ | 15 | 20 | 25 |
| y | 34.1 | 45.1 | 56.1 |

| | $X_1$ | $X_2$ | $X_3$ | $X_4$ | $X_5$ | $X_6$ | $X_7$ | $X_8$ |
|---|---|---|---|---|---|---|---|---|
| $X_1$ | 1 | 2 | 3 | 4 | 8 | 9 | 10 | 60 |
| $X_2$ | 1 | 3 | 5 | 6 | 16 | 7 | 5 | 3 |
| y | 2 | 4 | 6 | 8 | 16 | 18 | 20 | 22 |

$$W = \begin{pmatrix} 1.77 \\ 0.32 \\ 0.99 \end{pmatrix}$$

$$\|W\|^2 = 4.2$$

| | $Test_1$ | $Test_2$ | $Test_3$ |
|---|---|---|---|
| $X_1$ | 30 | 40 | 50 |
| $X_2$ | 15 | 20 | 25 |
| y | 26.2 | 34.4 | 42.5 |

# Adding Ridge Regularizer to the model λ=10



**Train**

| | $X_1$ | $X_2$ | $X_3$ | $X_4$ | $X_5$ | $X_6$ | $X_7$ | $X_8$ |
|---|---|---|---|---|---|---|---|---|
| $X_1$ | 1 | 2 | 3 | 4 | 8 | 9 | 10 | 11 |
| $X_2$ | 1 | 3 | 5 | 6 | 16 | 7 | 5 | 3 |
| y | 2 | 4 | 6 | 8 | 16 | 18 | 20 | 22 |

$$\underline{W} = \begin{vmatrix} 0.13 \\ 1.87 \\ 0.09 \end{vmatrix}$$

$$\|W\|^2 = 3.5$$

**Test**

| | $Test_1$ | $Test_2$ | $Test_3$ |
|---|---|---|---|
| $X_1$ | 30 | 40 | 50 |
| $X_2$ | 15 | 20 | 25 |
| y | 57.4 | 76.5 | 95.6 |

| | $X_1$ | $X_2$ | $X_3$ | $X_4$ | $X_5$ | $X_6$ | $X_7$ | $X_8$ |
|---|---|---|---|---|---|---|---|---|
| $X_1$ | 1 | 2 | 3 | 4 | 8 | 9 | 10 | 30 |
| $X_2$ | 1 | 3 | 5 | 6 | 16 | 7 | 5 | 3 |
| y | 2 | 4 | 30 | 8 | 16 | 18 | 20 | 22 |

$$\underline{W} = \begin{vmatrix} 0.54 \\ 0.73 \\ 0.81 \end{vmatrix}$$

$$\|W\|^2 = 1.48$$

| | $Test_1$ | $Test_2$ | $Test_3$ |
|---|---|---|---|
| $X_1$ | 30 | 40 | 50 |
| $X_2$ | 15 | 20 | 25 |
| y | 34.6 | 46.4 | 57.4 |

| | $X_1$ | $X_2$ | $X_3$ | $X_4$ | $X_5$ | $X_6$ | $X_7$ | $X_8$ |
|---|---|---|---|---|---|---|---|---|
| $X_1$ | 1 | 2 | 3 | 4 | 8 | 9 | 10 | 60 |
| $X_2$ | 1 | 3 | 5 | 6 | 16 | 7 | 5 | 3 |
| y | 2 | 4 | 60 | 8 | 16 | 18 | 20 | 22 |

$$\underline{W} = \begin{vmatrix} 0.82 \\ 0.33 \\ 1.1 \end{vmatrix}$$

$$\|W\|^2 = 1.92$$

| | $Test_1$ | $Test_2$ | $Test_3$ |
|---|---|---|---|
| $X_1$ | 30 | 40 | 50 |
| $X_2$ | 15 | 20 | 25 |
| y | 26.8 | 35.4 | 44.1 |

*pending recreation*

# [2] Ridge Regularization to overcome

# Inverse Problem of Singular Matrix

# Inverse Problem of Singular Matrix

$X^T X$

| | | |
|---|---|---|
| 14 | 10 | 15 |
| 10 | 40 | 46 |
| 20 | 80 | 92 |

$|X^T X| = 0$ ⇒ Singular Matrix ⇒ *No Inverse*

$(X^T X + \lambda I)$ $\lambda = 3$

| | | |
|---|---|---|
| 17 | 10 | 15 |
| 10 | 43 | 46 |
| 20 | 80 | 95 |

$|X^T X| = 5685$ ⇒ Non-Singular Matrix ⇒ *Inverse Exists*

$(X^T X + \lambda I)^{-1}$ ⇒

| | | |
|---|---|---|
| 0.07 | 0.04 | -0.03 |
| -0.01 | 0.23 | -0.11 |
| -0.01 | -0.2 | 0.11 |

# [3] Inverse of High Dimension (non-singular) Matrix

# Inverse of High Dimension Matrix

*Recall*

$$\underline{W} = (X^T X + \lambda I)^{-1} X^T \underline{Y}$$

if Feature vectors are of dimension d $\Rightarrow$ ($\underline{X}^T X$) is of size d+1 x d+1

The standard matrix inversion algorithm has a time complexity of $O(n^3)$, which means that the computation time increases rapidly with the size of the matrix (increasing cubically with the number of components).

For Example: if Features vector is of size 1000 then Inverse calculation is or order 1000,000,000 which is impractical

**Solution:**

*Instead of using **Least Squares** with Matrix inversion,*
*use **Gradient Descent** iterative solution*

# Gradient Descent (GD)

## and

## How to apply to minimize Loss function

*pending recreation*

# Gradient Descent Solution



**Gradient Descent (GD) Algorithm:**

[1] Start with initial **Random Weights (W)**

[2] Update the weights by moving along the direction of **negative gradient** of Loss function (w.r.t. W) (**moving step is a hyper parameter, AKA Learning Rate**)

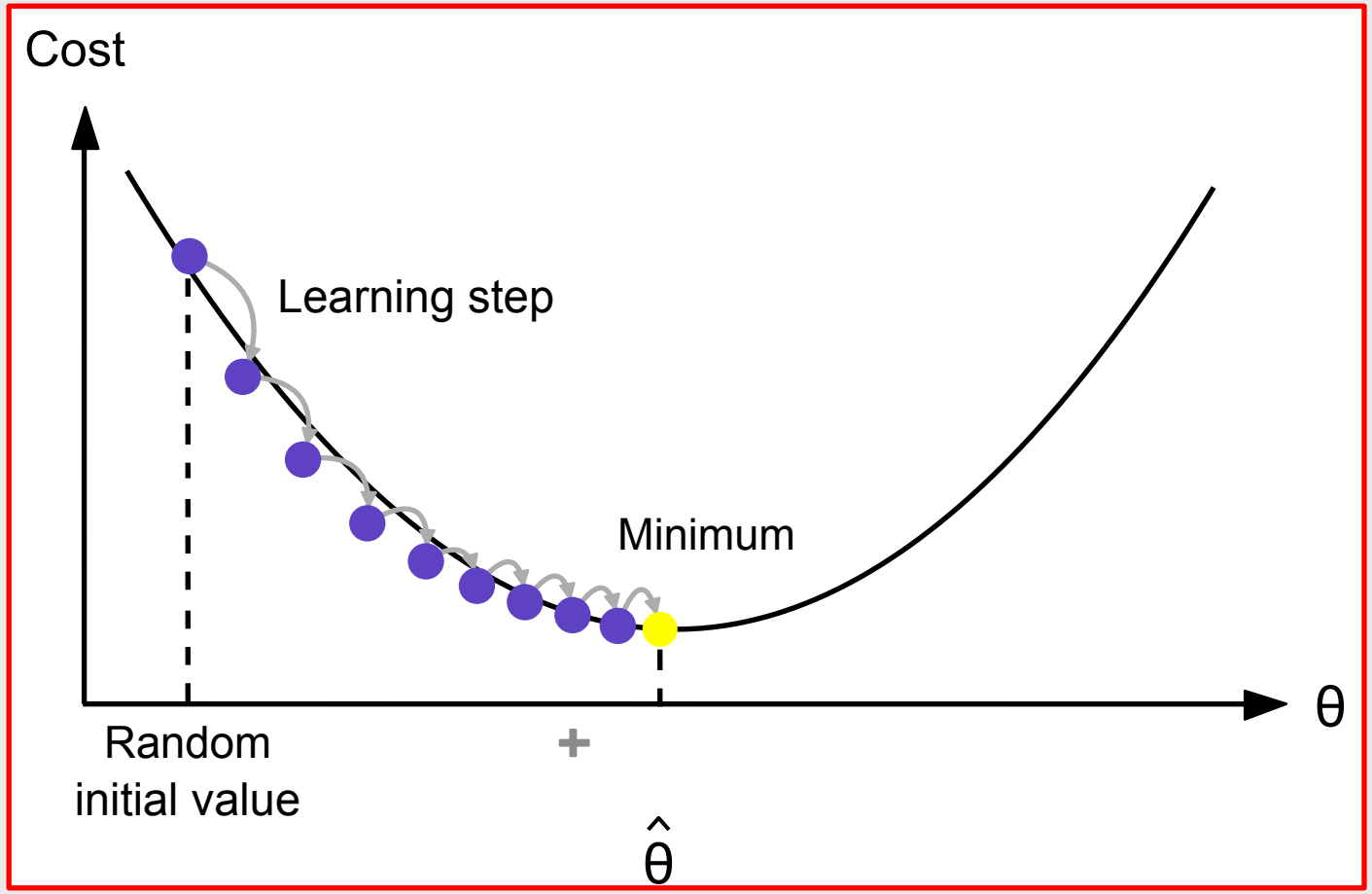
[3] Stop when the gradient is **almost zero** (this is at the minimum value of the function)

**Notes:**

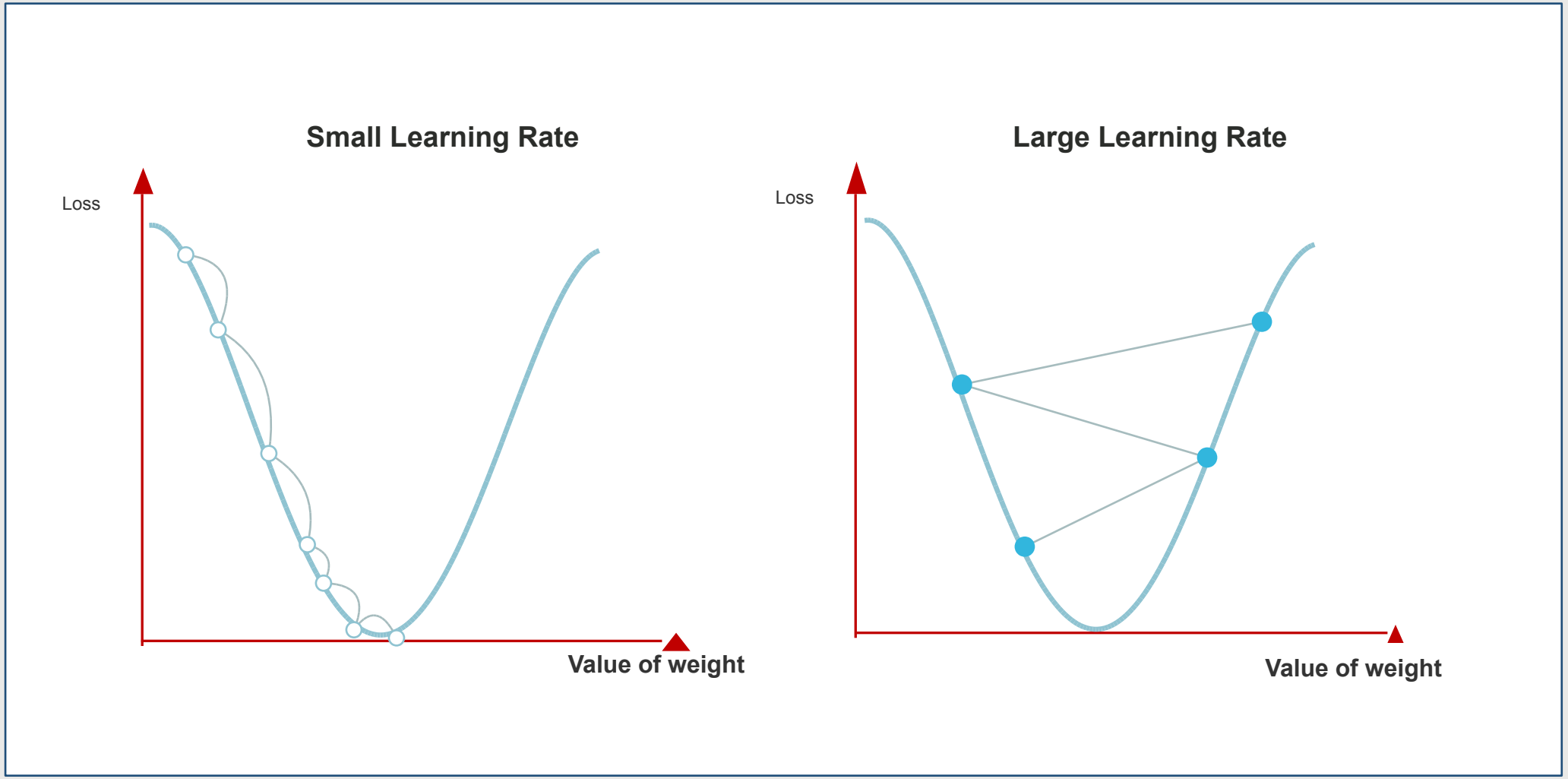
- **Least Squares** Linear regression objective function **is a convex function** (even with regularizer). This means it has **ONE Minimum** (good news ).

- GD will approximatly find this minimum value but at a variable number of steps depending on the used **Learning Rate**

# Follow Negative Direction of Gradient with Selected Learning Rate

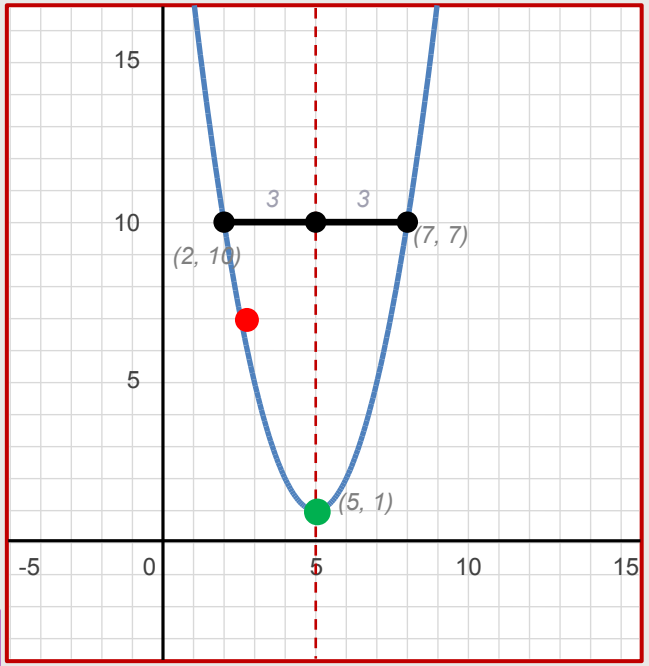


# Effect of small and large Learning Rate



# Example: Obtain Minimum Value of a function using Gradient Descent (GD)

# Convex Function, Learning Rate =0.2, Start at X=3




| step | 0.2 |
|---|---|
| initial X | 3 |
| delta | 0.1 |

$f(x) = (x-5)^2 + 1$

$X = 5$

*f(x)* Minimum value is **1.0**

***Reach Minimum after 11 Steps***

| x | 3 | 3.2 | 3.4 | 3.6 | 3.8 | 4 | 4.2 | 4.4 | 4.6 | 4.8 | 5 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| f(x) | 5 | 4.24 | 3.56 | 2.96 | 2.44 | 2 | 1.64 | 1.36 | 1.16 | 1.04 | 1 |
| x-delta | 2.9 | 3.1 | 3.3 | 3.5 | 3.7 | 3.9 | 4.1 | 4.3 | 4.5 | 4.7 | 4.9 |
| f(x-delta) | 5.41 | 4.61 | 3.89 | 3.25 | 2.69 | 2.21 | 1.81 | 1.49 | 1.25 | 1.09 | 1.01 |
| x+delta | 3.1 | 3.3 | 3.5 | 3.7 | 3.9 | 4.1 | 4.3 | 4.5 | 4.7 | 4.9 | 5.1 |
| f(x+delta) | 4.61 | 3.89 | 3.25 | 2.69 | 2.21 | 1.81 | 1.49 | 1.25 | 1.09 | 1.01 | 1.01 |
| slope | -8 | -7.2 | -6.4 | -5.6 | -4.8 | -4 | -3.2 | -2.4 | -1.6 | -0.8 | 0 |
| next x | 3.2 | 3.4 | 3.6 | 3.8 | 4 | 4.2 | 4.4 | 4.6 | 4.8 | 5 | stop |

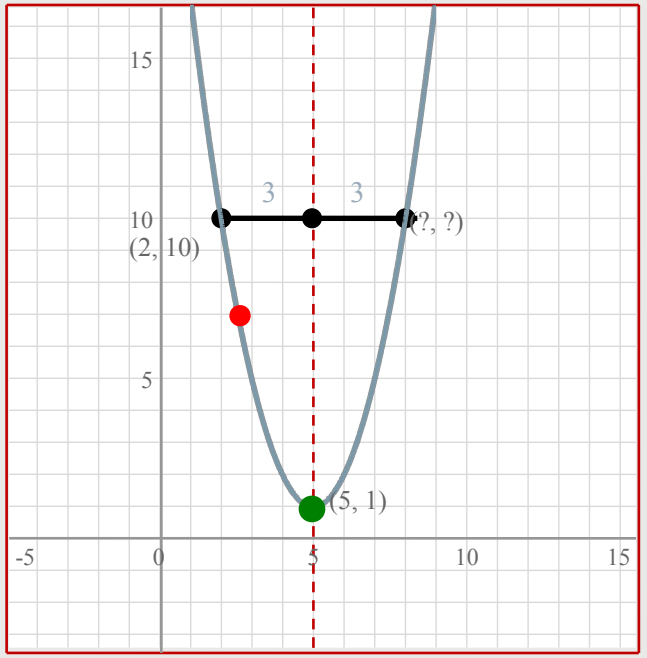
*pending recreation*

# Convex Function, Learning Rate =0.4, Start at X=3

| step | 0.4 |
|---|---|
| initial X | 3 |
| delta | 0.1 |

$$f(x) = (x-5)^2 + 1$$

***X* = 5**
*f(x)* Minimum value is **1.0**




***Reach Minimum after 6 Steps***

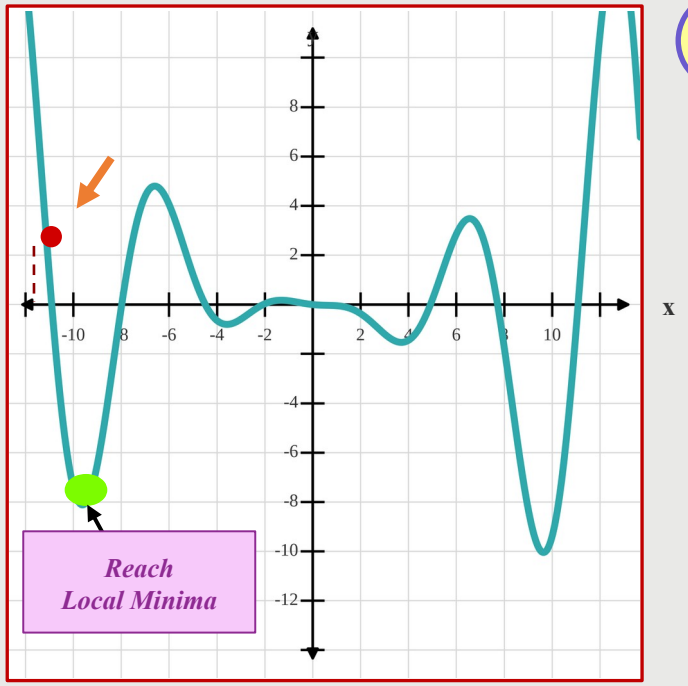
| x | 3 | 3.4 | 3.8 | 4.2 | 4.6 | 5 |
|---|---|---|---|---|---|---|
| f(x) | 5 | 3.56 | 2.44 | 1.64 | 1.16 | 1 |
| x-delta | 2.9 | 3.3 | 3.7 | 4.1 | 4.5 | 4.9 |
| f(x-delta) | 5.41 | 3.89 | 2.69 | 1.81 | 1.25 | 1.01 |
| x+delta | 3.1 | 3.5 | 3.9 | 4.3 | 4.7 | 5.1 |
| f(x+delta) | 4.61 | 3.25 | 2.21 | 1.49 | 1.09 | 1.01 |
| slope | -8 | -6.4 | -4.8 | -3.2 | -1.6 | 0 |
| next x | 3.4 | 3.8 | 4.2 | 4.6 | 5 | stop |

# Non-Convex Function

## May reach different minimum value based on:

## initial point and learning rate

$$f(x) = \frac{x^2\cos(x) - x}{10}$$




| step | 0.1 |
|---|---|
| initial X | -11 |
| delta | 0.1 |

*X* is in between -9.7 and color red bold -9.6

*f(x)* is in between -8.08 and -8.011

| x | -11 | -10.9 | -10.8 | -10.7 | -10.6 | -10.5 | -10.4 | -10.3 | -10.2 | -10.1 | -10 | -9.9 | -9.8 | -9.7 | -9.6 | -9.7 | -9.6 | -9.7 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| f(x) | 1.15 | -0.04 | -1.19 | -2.26 | -3.27 | -4.19 | -5.03 | -5.77 | -6.41 | -6.95 | -7.39 | -7.72 | -7.96 | -8.08 | -8.11 | -8.08 | -8.11 | -8.08 |
| x-delta | -11.1 | -11 | -10.9 | -10.8 | -10.7 | -10.6 | -10.5 | -10.4 | -10.3 | -10.2 | -10.1 | -10 | -9.9 | -9.8 | -9.7 | -9.8 | -9.7 | -9.8 |
| f(x-delta) | 2.39 | 1.15 | -0.04 | -1.19 | -2.26 | -3.27 | -4.19 | -5.03 | -5.77 | -6.41 | -6.95 | -7.39 | -7.72 | -7.96 | -8.08 | -7.96 | -8.08 | -7.96 |
| x+delta | -10.9 | -10.8 | -10.7 | -10.6 | -10.5 | -10.4 | -10.3 | -10.2 | -10.1 | -10 | -9.9 | -9.8 | -9.7 | -9.6 | -9.5 | -9.6 | -9.5 | -9.6 |
| f(x+delta) | -0.04 | -1.19 | -2.26 | -3.27 | -4.19 | -5.03 | -5.77 | -6.41 | -6.95 | -7.39 | -7.72 | -7.96 | -8.08 | -8.11 | -8.05 | -8.11 | -8.05 | -8.11 |
| slope | -24.3 | -23.4 | -22.2 | -20.8 | -19.3 | -17.6 | -15.8 | -13.8 | -11.8 | -9.8 | -7.7 | -5.7 | -3.6 | -1.5 | 0.3 | -1.5 | 0.3 | -1.5 |
| next x | -10.9 | -10.8 | -10.7 | -10.6 | -10.5 | -10.4 | -10.3 | -10.2 | -10.1 | -10 | -9.9 | -9.8 | -9.7 | -9.6 | -9.7 | -9.6 | -9.7 | -9.6 |

*pending recreation*

# Using Gradient Descent to train Linear Regression Model

## (Problem Formulation)

# Problem Formulation



Define two classes: class I and Class II

let $d(X)$ be a discriminant function = $W^T X$

if $X \in$ class I then $d(X) > 0$

if $X \in$ Class II then $d(X) < 0$

For misclassified $X$ :

If $X \in$ class I and misclassified as Class II then $d(X)$ will be $<0$

If $X \in$ Class II and misclassified as Class I then $d(X)$ will be $>0$

Define Error function as:

$$\text{Error} = \sum_{\text{all misclassified vectors}(i)} W^T X_i \quad \Rightarrow \quad (W^T X_i) \text{ take positive and negative values}$$

X

Define $\delta_i$ as :

If $X \in$ class I and misclassified as Class II then $d(X)$ will be $<0 \Rightarrow \delta_i = -1 \Rightarrow d(X)\,\delta_i > 0$

If $X \in$ Class II and misclassified as Class I then $d(X)$ will be $>0 \Rightarrow \delta_i = +1 \Rightarrow d(X)\,\delta_i > 0$

# Problem Formulation



**Define $\delta_i$ as :**

*If* $\underline{X} \in$ ***class I*** and ***misclassified*** as ***Class II*** *then* $d(\underline{X})$ *will be* $<0 \Rightarrow \delta_i = -1 \Rightarrow d(\underline{X})\delta_i > 0$

*If* $\underline{X} \in$ ***class II*** and ***misclassified*** as ***Class I*** *then* $d(\underline{X})$ *will be* $>0 \Rightarrow \delta_i = +1 \Rightarrow d(\underline{X})\delta_i > 0$

**Redefine Error function as:**

$$\text{Error} = \sum_{\text{all misclassified vectors } (i)} \delta_i \underline{W}^T \underline{X}_i \Rightarrow (\underline{W}^T \underline{X}_i) \text{ takes positive and negative values and } \delta_i \text{ has the same sign}$$

Summation is always ***positive***

*Minimize the Error using* ***Gradient Descent*** *with step* $\eta$

$$\underline{W}^{(updated)} = \underline{W}^{old} - \eta \frac{\partial Error}{\partial W} \quad \textbf{where} \quad \frac{\partial Error}{\partial W} = \sum_{\textit{misclassified} \text{ vectors}} \delta_i \underline{X}_i$$

$$\underline{W}^{(updated)} = \underline{W}^{old} - \eta \sum_{\textit{misclassified} \text{ vectors } \textit{ONLY} \text{ (i)}} \delta_i \underline{X}_i$$

# Gradient Descent (GD)

# vs

# Stochastic Gradient Descent (SGD)

# Gradient Descent (GD) – Stochastic Gradient Descent SGD) and Mini-Batch Gradient Descent

- Stochastic Gradient Descent (SGD) is a variant of GD which computes the gradient of the Error function with respect to a single training example
- In other words, it updates the weights using one training sample at a time
- Remember: GD updated the weight vector after passing through ALL training samples.

$$W^{(updated)} = W^{old} - \eta \delta_i X_i$$ Repeat for ALL misclassified Samples

**Why using SGD**

- SGD is usually more efficient than GD
- SGD converges faster than GD

**Batch GD is a generalization of GD and SGD where weights are updated after passing through "Mini-Batch" of training samples with size n**

**If n=N (it is GD), and if n=1 (it is SGD)**

# Training of Linear Regression Model

## usnig SGD

## Solved Example

$x_1 \; x_2 \; x_3 \; x_4 \; x_5 \; x_6$

$\begin{pmatrix}1\\1\end{pmatrix} \begin{pmatrix}2\\1\end{pmatrix} \begin{pmatrix}1\\2\end{pmatrix} \begin{pmatrix}5\\4\end{pmatrix} \begin{pmatrix}6\\5\end{pmatrix} \begin{pmatrix}6\\6\end{pmatrix}$

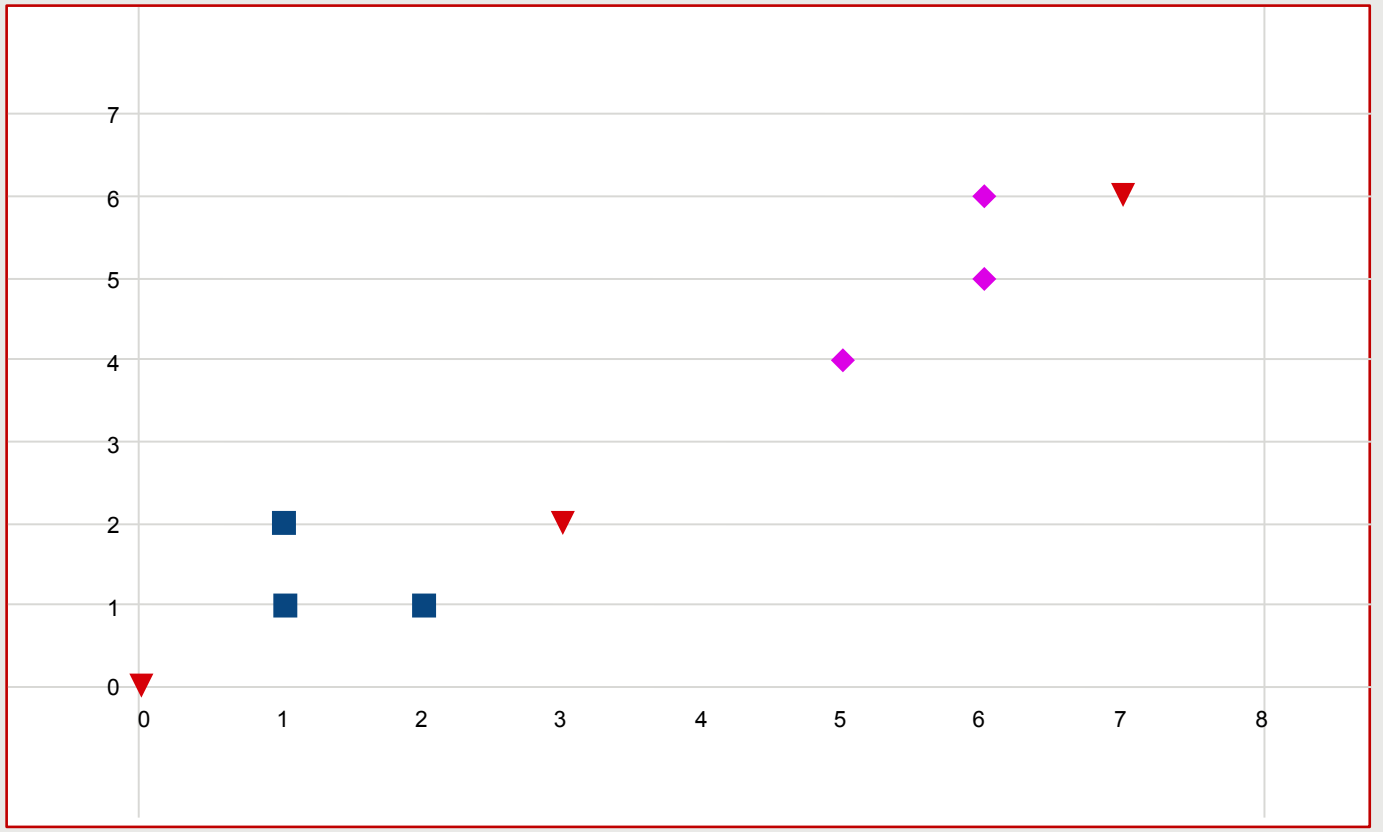
$c_1 \; c_1 \; c_1 \; c_2 \; c_2 \; c_2$

Initial W = $\begin{pmatrix}4\\-2\\-2\end{pmatrix}$ *Step η = 2*

$C_1$ positive, $C_2$ negative, Zero is unaccepted

Check classes of $z_1 \; z_1 \; z_1$

$\begin{pmatrix}0\\0\end{pmatrix} \begin{pmatrix}7\\6\end{pmatrix} \begin{pmatrix}3\\2\end{pmatrix}$



*pending recreation*

*pending recreation*

*pending recreation*

| $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $x_6$ |
|---|---|---|---|---|---|
| (1) | (2) | (1) | (5) | (6) | (6) |
| (1) | (1) | (2) | (4) | (5) | (6) |
| (1) | (1) | (1) | (4) | (5) | (6) |
| $C_1$ | $C_1$ | $C_1$ | $C_2$ | $C_2$ | $C_2$ |

Initial $\underline{W} = \begin{vmatrix} 4 \\ -2 \\ -2 \end{vmatrix}$ ***Step η = 2***

$C_1$ positive, $C_2$ negative, **Zero** is unaccepted

Check classes of

| $Z_1$ | $Z_1$ | $Z_1$ |
|---|---|---|
| (0) | (7) | (3) |
| (0) | (6) | (2) |

*Ugmented Vectors ( from Top )*

| | | | | | |
|---|---|---|---|---|---|
| (1) | (1) | (1) | (1) | (1) | (1) |
| (1) | (2) | (1) | (5) | (6) | (6) |
| (1) | (1) | (2) | (4) | (5) | (6) |
| (1) | (1) | (1) | (4) | (5) | (6) |
| $C_1$ | $C_1$ | $C_1$ | $C_2$ | $C_2$ | $C_2$ |

**Iteration 4**

**No change in $\underline{W}$**

***Stop Iterations***

$d(x_1) = 1\ 1\ 1 \begin{vmatrix} 14 \\ -4 \\ 0 \end{vmatrix} = 10$ ⇒ Keep same Weight $\underline{W} = \begin{vmatrix} 14 \\ -4 \\ 0 \end{vmatrix}$

$d(x_2) = 1\ 2\ 1 \begin{vmatrix} 14 \\ -4 \\ 0 \end{vmatrix} = 6$ ⇒ Keep same Weight $\underline{W} = \begin{vmatrix} 14 \\ -4 \\ 0 \end{vmatrix}$

$d(x_3) = 1\ 1\ 2 \begin{vmatrix} 14 \\ -4 \\ 0 \end{vmatrix} = 10$ ⇒ Keep same Weight $\underline{W} = \begin{vmatrix} 14 \\ -4 \\ 0 \end{vmatrix}$

$d(x_4) = 1\ 5\ 4 \begin{vmatrix} 14 \\ -4 \\ 0 \end{vmatrix} = -6$ ⇒ Keep same Weight $\underline{W} = \begin{vmatrix} 14 \\ -4 \\ 0 \end{vmatrix}$

$d(x_5) = 1\ 6\ 5 \begin{vmatrix} 14 \\ -4 \\ 0 \end{vmatrix} = -10$ ⇒ Keep same Weight $\underline{W} = \begin{vmatrix} 14 \\ -4 \\ 0 \end{vmatrix}$

$d(x_6) = 1\ 6\ 6 \begin{vmatrix} 14 \\ -4 \\ 0 \end{vmatrix} = -112$ ⇒ Keep same Weight $\underline{W} = \begin{vmatrix} 14 \\ -4 \\ 0 \end{vmatrix}$

$x_1$ $x_2$ $x_3$ $x_4$ $x_5$ $x_6$

$\begin{pmatrix}1\\1\end{pmatrix}$ $\begin{pmatrix}2\\1\end{pmatrix}$ $\begin{pmatrix}1\\2\end{pmatrix}$ $\begin{pmatrix}5\\4\end{pmatrix}$ $\begin{pmatrix}6\\5\end{pmatrix}$ $\begin{pmatrix}6\\6\end{pmatrix}$

$C_1$ $C_1$ $C_1$ $C_2$ $C_2$ $C_2$

Initial $\underline{W} = \begin{pmatrix}4\\-2\\-2\end{pmatrix}$ ***Step η = 2***

$C_1$ *positive* $C_2$ *negative* ***Zero*** *is unaccepted*

Check classes of $Z_1$ $Z_1$ $Z_1$

$\begin{pmatrix}0\\0\end{pmatrix}$ $\begin{pmatrix}7\\6\end{pmatrix}$ $\begin{pmatrix}3\\2\end{pmatrix}$

$\underline{W} = \begin{pmatrix}14\\-4\\0\end{pmatrix}$

***Ugmented Vectors ( from Top )***

$x_1$ $x_2$ $x_3$ $x_4$ $x_5$ $x_6$

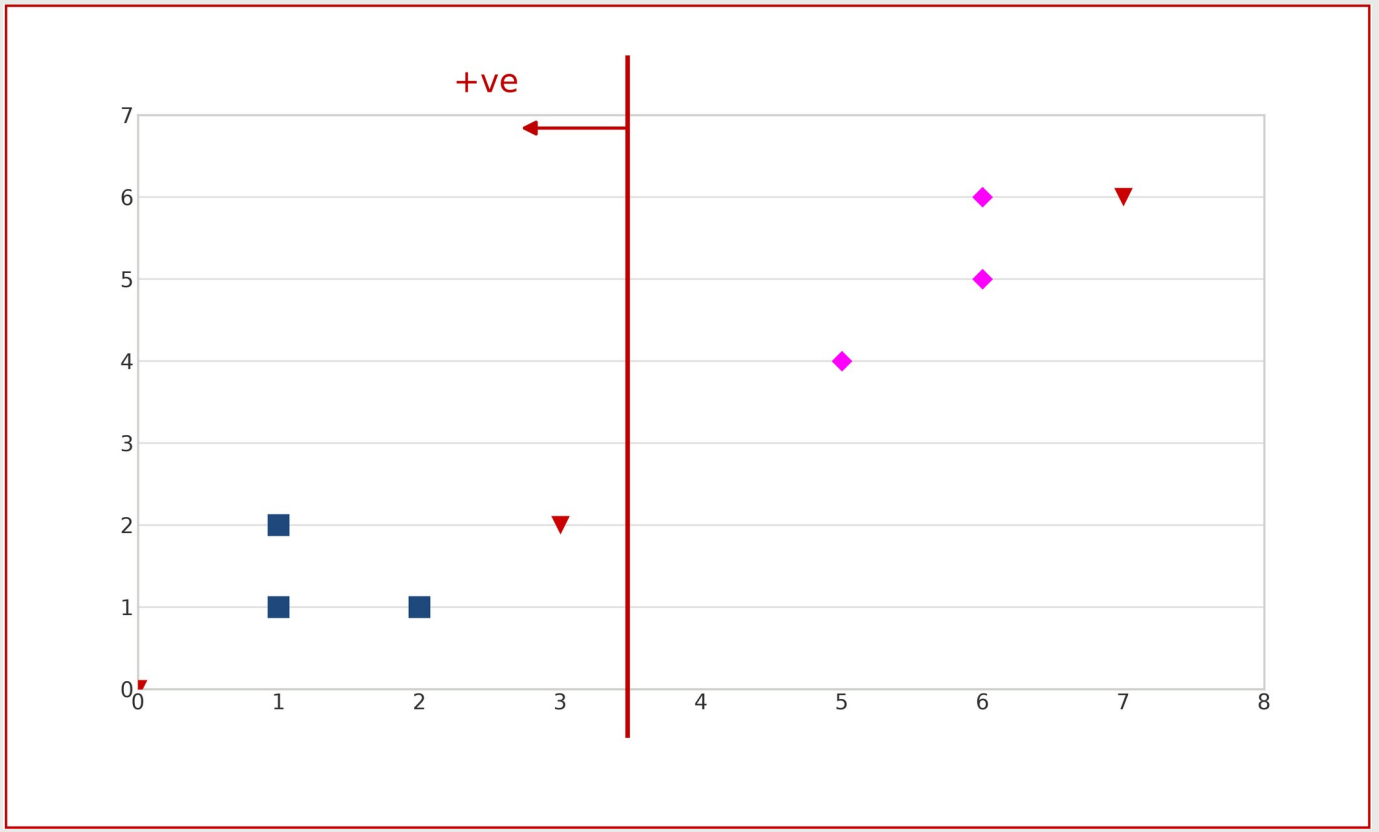
$\begin{pmatrix}1\\1\\1\end{pmatrix}$ $\begin{pmatrix}1\\2\\1\end{pmatrix}$ $\begin{pmatrix}1\\1\\2\end{pmatrix}$ $\begin{pmatrix}1\\5\\4\end{pmatrix}$ $\begin{pmatrix}1\\6\\5\end{pmatrix}$ $\begin{pmatrix}1\\6\\6\end{pmatrix}$

$C_1$ $C_1$ $C_1$ $C_2$ $C_2$ $C_2$

## Iteration 4

No change in $\underline{W}$

***Stop*** *Iterations*



$x_1$ $x_2$ $x_3$ $x_4$ $x_5$ $x_6$

$\begin{pmatrix}1\\1\end{pmatrix}$ $\begin{pmatrix}2\\1\end{pmatrix}$ $\begin{pmatrix}1\\2\end{pmatrix}$ $\begin{pmatrix}5\\4\end{pmatrix}$ $\begin{pmatrix}6\\5\end{pmatrix}$ $\begin{pmatrix}6\\6\end{pmatrix}$

$C_1$ $C_1$ $C_1$ $C_2$ $C_2$ $C_2$

Initial $\underline{\mathbf{W}} = \begin{pmatrix}4\\-2\\-2\end{pmatrix}$ ***Step η = 2***

$C_1$ positive $C_2$negative Zero is unaccepted

Check classes of $Z_1$ $Z_1$ $Z_1$

$\begin{pmatrix}0\\0\end{pmatrix}$ $\begin{pmatrix}7\\6\end{pmatrix}$ $\begin{pmatrix}3\\2\end{pmatrix}$

*Ugmented Vectors ( from Top )*

$x_1$ $x_2$ $x_3$ $x_4$ $x_5$ $x_6$

$\begin{pmatrix}1\\1\\1\end{pmatrix}$ $\begin{pmatrix}1\\2\\1\end{pmatrix}$ $\begin{pmatrix}1\\1\\2\end{pmatrix}$ $\begin{pmatrix}1\\5\\4\end{pmatrix}$ $\begin{pmatrix}1\\6\\5\end{pmatrix}$ $\begin{pmatrix}1\\6\\6\end{pmatrix}$

$C_1$ $C_1$ $C_1$ $C_2$ $C_2$ $C_2$

## Check Classes

$d(Z_1) = (1\ 0\ 0)\begin{pmatrix}14\\-4\\0\end{pmatrix} = 14$ ⇒ **Positive > 0** **Class I**

$d(Z_2) = (1\ 7\ 6)\begin{pmatrix}14\\-4\\0\end{pmatrix} = -14$ ⇒ **Negative < 0** **Class II**

$d(Z_2) = (1\ 3\ 2)\begin{pmatrix}14\\-4\\0\end{pmatrix} = -2$ ⇒ **Negative < 0** **Class II**

*pending recreation*

*pending recreation*

*pending recreation*

*pending recreation*